

**Seat
No.**

Day & Date: Wednesday, 29-04-2026
Time: 12:00 PM To 01:30 PM

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.
3) Use of scientific calculator is allowed.

06

- 1) If $X_0 = 3$, $f(X_0) = 24$ $X_1 = 5$, $f(X_1) = 50$ then value of first order divided difference $f(X_0, X_1) = [X_0, X_1]$ is _____.
a) 13 b) 26
c) -13 d) -26
- 2) The square matrix in which all $a_{ij} = 0$; for all $i < j$ is called as _____ matrix.
a) lower triangular b) upper triangular
c) symmetric d) diagonal
- 3) Interpolation means estimating numerical value at a point which lies _____.
a) outside the given range of arguments
b) inside the given range of arguments
c) outside the range of the dependent variable
d) None of these
- 4) $EV = VE =$ _____.
a) V b) E^{-1}
c) E d) Δ
- 5) While doing multiplication of two numbers in normalized floating point notation mantissas should be _____ and exponents should be _____ respectively.
a) added and multiplied b) multiplied and added
c) added and made equal d) made equal and added
- 6) One of root of the equation lies in between X_0 and X_1 if and only if $f(X_0)$ and $f(X_1)$ have _____.
a) same signs b) positive signs
c) negative signs d) opposite signs

Q.2 Answer the following questions. (Any Three)

06

- a) Define forward difference operator Δ and backward difference operator ∇ .
- b) State Regula Falsi method formula to find first approximation for the value of root of equation $f(X) = 0$ in the interval $[X_0, X_1]$.
- c) Prepare forward difference table for the following data.
- | | | | | |
|------------|---------|---------|---------|---------|
| X | 5 | 10 | 15 | 20 |
| $Y = F(x)$ | 22.6589 | 38.2456 | 49.2589 | 65.2134 |
- d) Define diagonal matrix and scalar matrix with suitable example.
- e) Solve
- $32.4589 E 25 \times 8.2541 E 20$
 - $125.52 E 320 \div 28.5485 E 120$
- Express your final answer in normal form.

Q.3 Answer the following questions. (Any Two)

06

- a) Evaluate $\left(\frac{\Delta^2}{E}\right)x^2$ by taking $h = 1$.
- b) Let $A = \begin{bmatrix} 2 & 3 \\ 0 & -5 \\ 5 & 1 \end{bmatrix}_{3 \times 2}$ and $B = \begin{bmatrix} -5 & 5 \\ 4 & 0 \\ 9 & -5 \end{bmatrix}_{3 \times 2}$
Find the matrix $2A + B$ and $A - 3B$
- c) By using bisection method find the approximate value of root of equation $f(x) = e^x - 4x$ in the interval $[1, 0]$. Perform only three iterations.

Q.4 Answer the following questions. (Any Two)

06

- a) Define absolute error, relative error and percentage error.
- b) Solve the following system of linear equations by using Jacobi's method. Perform only two iterations.
 $6X_1 + X_2 + X_3 = 20$; $X_1 + 4X_2 - X_3 = 6$; $X_1 - X_2 + 5X_3 = 7$
- c) By using Lagrange's Interpolation Formula, find numerical value of $f(9.5)$ from the data given below.

X	5	8	10
$Y = f(x)$	20.2345	35.9876	58.4567

Q.5 Answer the following question. (Any One)

06

- a) By using Newton's Forward difference interpolation formula find numerical value of y at $x = 3.5$ from the following data.

X	2	4	6	8
$Y = F(x)$	112.26	128.96	149.45	190.52

- b) Solve the following system of linear equations by using Gauss Seidel iterative method. Perform only three iterations.
 $x + 3y + 10z = 24$; $28x + 4y - z = 32$; $2x + 17y + 4z = 35$